Mawlana Bhashani Science and Technology University
Department of Information and Communication Technology
4th Year 1st Semester B.Sc (Engg.) Final Examination-2025

Course Title: Optical Communication Course Code: **ICT-4105**

Full Marks: 70 Times: 3 hours

[Answer any five from the following questions. Illustrate where necessary.]

1 (a) Explain the basic block diagram of a fiber-optic communication system. Discuss the functions of the transmitter, optical fiber, receiver, and associated components. **05**

(b) What is sampling? Why is the standard 4-kHz telephone channel sampled 8000 times a second? **04**

(c) A light emitting diode radiates 2mW. Compute the dBm value of this radiated power. The power travels through a group of components having a combined loss of 23 dB. Compute the output power. **05**

2 (a) Discuss the advantages of optical fibers over conventional coaxial cables. **04**

(b) An optical fiber has a core refractive index of 1.48 and cladding refractive index of 1.46. Calculate: **06**

i. Numerical aperture
ii. Acceptance angle in air
iii. Critical angle at the core-cladding interface.

(c) What is the concept of total internal reflection? Explain with a suitable cable setup. **04**

3 (a) What is photonics? ~~Draw the block diagram of the fundamental's optical fiber communication and explain the function of each block.~~ **04**

(b) What is the basic difference between TE and TM polarization? Write Fresnel's equation for TE and TM polarization. **05**

(c) For the plastic fiber, $n_1 = 1.495$ and $n_2 = 1.402$. Calculate the acceptance angle and numerical aperture. **05**

(d) The input power to an optical fiber is 2 mW while the power measured at the output end is 200 μW. If the fiber attenuation is 0.4 *dB/km* at 1310 nm, calculate the length of the fiber. **04**

4 (a) With suitable diagrams, explain the structure of step-index and graded-index optical fibers. Compare their characteristics. **04**

(b) A step-index fiber has a core diameter of 50 μm, core refractive index of 1.50, cladding refractive index of 1.47, and operates at 850 nm. Determine the normalized frequency and estimate the number of guided modes. **06**

(c) Multimode GRIN fiber produces less modal pulse distortion than SI fiber. Why? **04**

5 (a) What is attenuation in an optical fiber? Explain the major mechanisms responsible for attenuation, including material absorption, Rayleigh scattering, and bending losses. **06**

(b) Explain modal dispersion, material dispersion, and waveguide dispersion. How do these dispersions affect the bandwidth of an optical fiber? **05**

(c) What is Brewster angle? What happens if light is incident in this angle? **03**

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6 (a) Draw the diagram of a typical double-heterojunction structure LED, along with energy band diagram and refractive index profile, hence explain the operation. Why is it advantageous compare to homojunction LED? **06**

(b) What is LASER? Explain spontaneous and stimulated emission with necessary sketch. **04**

(c) What is the importance of optical amplifier? Describe the working principle of EDFA. **04**

7 (a) Draw and explain the complete block diagram of a practical fiber-optic communication link. **06**

(b) A digital optical communication system operates at 1550 nm over a 40-km fiber link. The fiber attenuation is 0.22 dB/km. Connector losses are 1 dB at each end, and there are four splices, each having a loss of 0.2 dB. If the transmitter output power is +3 dBm and the receiver sensitivity is −28 dBm, calculate the system power margin. **06**

(c) What are basic requirements of fiber optic material. **02**

8 (a) With basic operational principle of WDM, explain the working of typical WDM network and mention WDM standards. **04**

(b) Write short note on Fiber Bragg Grating. **05**

(c) Design a digital system to transmit a 400 Mbps NRZ signal over 100 km path with an error rate of 10^{-9} or better. **05**

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